

Delphine Doutsas

PhD Student at Mines Paris - PSL University

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I am currently a third-year PhD student at Mines Paris - PSL and École Normale Supérieure, under the supervision of Bruno Figliuzzi and Peter Neri. I am broadly interested in inverse problems, with my current work focusing more specifically on simplex-constrained inverse problems, especially in settings where observations can be modeled as mixtures or convex combinations of underlying components. My research initially focused on hyperspectral unmixing and has progressively broadened toward generative modeling, structured latent representations, and model explainability.

Education

Mines Paris, Ecole Normale Supérieure - PSL University <i>PhD in Applied Mathematics</i> Applied mathematics to address the problem of blind hyperspectral unmixing in challenging scenarios.	Paris, France 2023–Currently
Sorbonne University - Paris 6 <i>MSc in Engineering of Robotics and Automatic Systems</i> High honors with ranking 5/51 (1st semester), 4/53 (2nd semester) and 6/67 (3rd semester). Master thesis graded 17/20.	Paris, France 2020–2022
Sorbonne University - Paris 6 <i>BSc in Mathematics (Gap year in 2017).</i>	Paris, France 2016–2020
Sorbonne University - Paris 6 <i>Additional courses in Physics</i> Courses taken in parallel with the degrees' courses (BSc level): Mechanics, Special Relativity, Introduction to Thermodynamics.	Paris, France

Experience

Thales Land And Air Systems <i>Apprenticeship during MSc 2nd year</i> Parallelization and optimization of radar signal processing filters for GPU. Unfortunately, confidential content.	Limours, France 2022
Sorbonne University - Paris 6 <i>Research Project: Introduction to Research</i> Recognition and reproduction of body poses by the PEPPER robot with programming on the meta-operating system ROS.	Paris, France 2022
Institut du Cerveau et de la Moelle Epinière (ICM) <i>Summer Research Internship</i> Deep Learning for melanoma characterization on histopathological images.	Paris, France 2021

Publications and pre-publications

- D. Doutsas, B. Figliuzzi, A Critical β -Scale for Posterior Collapse in Dirichlet β -VAEs, [Currently under review at NIPS 2026]
- D. Doutsas, B. Figliuzzi, Effective-dimension control of β in Dirichlet β -VAE for Blind Hyperspectral Unmixing, CVPR, MORSE workshop, 2026
- D. Doutsas, B. Figliuzzi, Optimal Transport Based Hyperspectral Unmixing for Highly Mixed Observations, IEEE WHISPERS conference, 2024

Conferences

- Jun 2025:** Poster. CVPR 2026 MORSE workshop, Denver (United States of America)
- Jun 2025:** Poster. The Beg Rohu Summer School of Physics, Saint-Pierre Quiberon (France)
- Apr 2025:** Presentation. French-German Workshop on Mathematical Image Processing, Fraunhofer ITWM, Kaiserslautern (Germany)
- Dec 2024:** Presentation. IEEE WHISPERS conference, Helsinki (Finland)

Event Participation

2026: IDRIS Open Hackathon. Participated in a GPU HPC & AI hackathon focused on accelerating and optimizing scientific and AI codes on modern GPU architectures. GitHub repository.

Teaching

Mines Paris - PSL University

2025 - now: IASD M.Sc - Deep Learning for Image Analysis

2024 - now: 1st year Engineering Program - Signal Processing and Functional Analysis.

2023 - 2024: Bioluminescence Imaging (BIM) M.Sc, Machine Learning

Sorbonne University

2021 - 2022: Tutoring for B.Sc students.

Supervision

Co-Advisor: Haidar Ali Youssef - Engineering student from Mines Paris on a 3-month research project, advised with B. Figliuzzi.

Awards and Distinctions

- Co-recipient of the *Best paper award* at CVPR 2026 MORSE workshop.
- Co-recipient of the *Bioucas-Dias outstanding paper award* at WHISPERS 2024 conference.

Skills

Languages: French (native), English (fluent), Spanish (basic skills)

Programming: Python (NumPy, Keras, Pytorch, Tensorflow), LaTeX, Git, CUDA language.